$$I_1 = U_g(R,z) + \frac{1}{2}\left(\dot{R}^2 + R^2\dot{\theta}^2 + \dot{z}^2\right)$$
$$\boxed{I_2 = R^2\dot{\theta}}$$

"A_3 is easily seen, if we introduce the function

$$U(R,z) = U_g(R,z) + \frac{c_2^2}{2R^2}, \quad c_2 = \text{const}$$

=> equation of motion: $\ddot{R} = -\frac{\partial U}{\partial R}$, $\ddot{z} = -\frac{\partial U}{\partial z}$, how so? proof!

-> leads to equation of motion on a plane => phase space $(x, y, \dot{x}, \dot{y})$

-> first conserving and isolating integral

$$I_1 = U(x,y) + \frac{1}{2}(\dot{x}^2 + \dot{y}^2)$$

I_2 can be obtained through Energy Integral -> $U(x,y) + \frac{1}{2}(\dot{x}^2 + \dot{y}^2) = E$

$$\dot{x}^2 \geq 0 \Rightarrow U(x,y) + \frac{1}{2}\dot{y}^2 \leq E$$

"if there is no other isolating integral the trajectory will fill the volume defined by $U(x,y) + \frac{\dot{y}^2}{2} \leq E$ [...] we shall call it ergodic"

- through looking at the crosssection at $x=0$, $\dot{x}>0$, i.e. let the system evolve and project the resulting positions onto that plane, if the points lie on ~~the pl~~ curve, there exists the second integral. Else the point would fill the entire volume.

=> phase space trajectories in $(y, \dot{y})$-plane suffice to lead to the understanding of the existence of I_2 - an isolating integral.